Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/22/2023
PI Name:	Ganguli, Gurudas Ph.D.		
Project Title:	Understanding the Frequency Synchronization Physics in PK-4 Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20375-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	04/15/2017	**End Date:**	12/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per U. Israelsson/JPL (Ed., 1/6/22) NOTE: End date changed to 10/15/2021 per U. Israelsson/JPL (Ed., 4/6/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crabtree, Chris Ph.D. (Naval Research Laboratory)		
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Performance Goal Text:			

Task Description:	The objective of this proposal is to provide theoretical support for a separate PK-4 (Plasma Krystall-4) experimental project, “Nonlinear Waves in Dusty Plasmas.” The experimental effort, led by Principal Investigator (PI) Goree, is an investigation of nonlinear synchronization, with the advance that it will study not point oscillators, but waves that propagate a distance of many wavelengths. The literature is lacking theory suitable for comparison to the experiment, perhaps because the experiments at this point are still very new. In particular, theory is needed for the case, beyond the point oscillator description that is most common, for synchronization in the realm of propagating waves. Beyond just supporting the PK-4 experiment, the theoretical advances we propose can have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the proposed investigation will be relevant to a broader area of contemporary interest and enhance the science foundation for nonlinear phenomena in plasma physics. A plasma system normally constitutes a highly complex medium that can display a rich variety of nonlinear phenomena arising out of a host of intrinsic oscillations and other non-stationary states. A clear identification of phenomena like synchronization or frequency entrainment in such a medium often poses serious experimental challenges. Many past studies on such systems have therefore relied on driving a single plasma system with an external frequency source. There have been very few studies devoted to the investigation of mutual synchronization between two plasma systems except where this phenomenon was studied between two double plasma devices and where it was investigated in two electrical discharges. There is no theoretical framework to account for these phenomena in a realistic dispersive plasma system that includes natural growth and dissipation of waves. Therefore, the primary focus of this proposal is to develop such a framework and apply it to understand the dedicated PK-4 dusty plasma experiment on frequency synchronization. The validated model will then be useful for general application to other experiments. Some of the science questions that we will address in our proposed theoretical investigation are the following: • What are the parametric domains in the state space of the external driver frequency and amplitude that lead to frequency entrainment, frequency pulling, and synchronization of the dust acoustic waves in the PK-4 plasma? • How do dispersion and nonlinearity of the dusty plasma medium influence the nature of the synchronization process? • Can the non-harmonic synchronized state, as observed in ground-based experiments, be predicted by a continuum model of the dusty plasma's constituents? • What determines the threshold amplitude for synchronization, and how is the threshold affected by dust-dust correlations (i.e., the strength of the Coulomb parameter)? • How does viscous dissipation in the medium affect synchronization and influence the nature of the synchronized excitation? • Can one observe phase-flip transitions in the synchronization, i.e., a sudden transition between in-phase synchronization to pi out-of-phase synchronization? • How does a broad-spectrum driver (e.g., signals from another plasma source) alter the synchronization scenario?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Beyond just supporting the PK-4 experiment, the theoretical advances in this project will have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the investigation is relevant to a broader area of contemporary interest.
Task Progress:	Synchronization Model Studies: In this year, we have completed a detailed numerical study of a model nonlinear system for studying the synchronization phenomenon in dusty plasma experiments aboard the (Plasma Krystall-4) PK-4 device. The study is based on the forced Korteweg-de Vries-Burgers (fKdV-B) equation, which is a generalization of the forced Korteweg-de Vries equation (fKdV) by incorporating dissipative effects arising from viscosity in the medium. This equation provides a proper theoretical framework and a better physical model compared to the Van der Pol oscillator model for studying the dynamics of nonlinear dust acoustic waves by properly accounting for nonlinear, dispersive and dissipative influences on the waves. Using the model, we have successfully demonstrated harmonic (1:1) and super-harmonic (1:2) synchronization states of dust acoustic waves (DAWs) for the experimental values reported in the laboratory experiment of Ruhunusiri et al. [W. D. S. Ruhunusiri and J. Goree, Phys. Rev. E 85, 516046401 (2012)]. In particular, comparison of our theoretical Arnold tongue diagram with their experimental one shows the following common features. As in the experimental Arnold tongue diagram, we see the existence of amplitude thresholds, as well as clear evidence of the branching phenomena. However, there are also important differences. With our model, we have not been able to obtain sub-harmonic synchronization that have been observed in the experiment. Furthermore, our model uses an external driver that closely resembles a nonlinear natural mode of the system; whereas, in the experiment, a purely time varying external sinusoidal driver has been used. However, it is not clear what form this driver takes inside the plasma system and whether it manifests itself as a spatio-temporally varying perturbation. These and other questions, such as the absence of sub-harmonic synchronization in the equation, the neglect of dissipation arising from gas friction on the dust particles, etc., remain to be explored in the future in order to further improve the model. This will be taken up in our future studies. Molecular Dynamic Simulations: In parallel to the above fluid model based study of synchronization, we are continuing with molecular dynamic (MD) simulation studies to provide a more realistic representation of the experimental dusty plasma system. Preliminary runs have been made using the open source LAMPPS code of 10,000 highly charged particles interacting through a Yukawa potential. Langevin dynamics models collisions with gas atoms, and thereby allows us to include dissipation in the system. Initial results show promising signatures of wave synchronization with an external driver and a host of diagnostic tests are being made to ascertain the numerical accuracy of the code and eliminate spurious physical effects. A concerted effort will be made in the coming year to conclude this study.

Bibliography Type:	Description: (Last Updated: 03/22/2023)
Articles in Peer-reviewed Journals	Mir A, Tiwari S, Sen A, Crabtree C, Ganguli G, Goree J. "Synchronization of dust acoustic waves in a forced Korteweg--de Vries--Burgers model." Phys. Rev. E. 2023 Mar 9;107(3):035202-9. https://link.aps.org/doi/10.1103/PhysRevE.107.035202 , Mar-2023